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Examining Decision Characteristics & Challenges for Agile Software Development

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Highlights

- In-depth case via interviews, focus groups, meeting observation & document analysis
- We conducted 18 individual interviews, two focus groups and observed 21 meetings
- Results identified particular decision characteristics across four key agile values
- Results show characteristics' decision process, intelligence and quality challenges
- This study provides a framework to evaluate agile decision making

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EXAMINING DECISION CHARACTERISTICS & CHALLENGES FOR AGILE SOFTWARE DEVELOPMENT

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Abstract

Although agile software development is often associated with improved decision making, existing studies tend to focus on narrow aspects of decision making in such environments. There is a lack of clarity on how teams make and evaluate a myriad of decisions from software feature inception to product delivery and refinement. Indeed there is relatively little known about a) the decision characteristics related to agile values, and b) the challenges they present for decision making on agile teams. We present an in-depth exploratory case study based on a pluralistic approach comprising semi-structured interviews, focus groups, team meeting observations, and document analysis. The study identifies failings of decision making in an agile setting. Explicitly considering the decision process, information intelligence used in decision making, and decision quality, the key contribution of this paper is the development of an over-arching framework of agile decision making, which identifies particular decision characteristics across 4 key agile values and the related challenges for agile team decision making. It provides a framework for researchers and practitioners to evaluate the decision challenges of an agile software development team and to improve decision quality.

Keywords: Agile software development; agile decision making; decision characteristics; decision process; decision intelligence; decision quality

1 INTRODUCTION

Agile software development (ASD) emerged in the late 1990s to address the uncertainty of customer requirements, technology evolution, and changing business environments. Agile approaches rejected the highly-formalised thinking of the time in favour of dynamic, user-centric, methods characterised by short development cycle iterations, continuous releases, and rapidly evolving requirements, dynamic underlying data, and reflection (Fowler and Highsmith, 2001; Schwaber and Beedle, 2002).

There are many characteristics of ASD that affect decision making, including its fast-paced iterative and incremental nature, its organic and flexible developer roles, and its emphasis on self-management (Austin and Devin, 2009; Moe and Aurum, 2008; Zannier et al., 2007; Zannier and Maurer, 2006; Zannier and Maurer, 2007). Differentiating features of decision making in ASD merit significant research. No research to date has attempted to identify in a holistic, structured manner, an over-arching set of ASD characteristics that affect decision making. This is concerning given that agility fundamentally affects software development decision making through a number of key characteristics; there is a need to understand the challenges to decision making that these over-arching ASD characteristics present (Henderson-Sellers and Serour, 2005). It is important to contextually research these characteristics and map them to challenges for decisions so that we can provide a framework for both researchers and practitioners to better evaluate decision making in ASD teams. Therefore the objectives of this research are to:

1. Identify contextually specific decision characteristics present in ASD.
2. Describe the challenges such characteristics pose for decision making for ASD teams.

We conducted an in-depth case study to examine these research objectives. This case study included 18 individual interviews, 2 focus groups, and observation of 21 team meetings with supplementary project documentation. We then used these data to create a framework of decision challenges for each decision characteristic identified.

In the sections that follow, we present a background to decision making and contextualise it for ASD. To provide a structure for the identification of decision characteristics and related challenges, we first discuss ASD, followed by decision processes, decision intelligence, and decision quality. We then outline our research method to describe our case study. Finally, we present our findings, discussion, and conclusions.

2 RELATED WORK

2.1 Agile Software Development

There are 2 dominant perspectives in the agility literature in terms of the enablement and evaluation of agility: (i) a method adherence-based view and (ii) a principle or value-based perspective. Many studies adopt the former, measuring agility by the number of XP or Scrum practices that are used (Dybå and Dingsøyr, 2008). However, there are 2 limitations of this narrow focus. First, agility is a vague and multi-faceted concept given that agile is practised in so many different ways: it is polymorphous, multi-dimensional, and highly exposed to diverse interpretation and application (Conboy, 2009). Second, methods are rarely, if ever, applied in textbook formats. Studying the specific practices of a specific method are less useful when we know it is unlikely that practices will be used in this original format. Therefore, this study takes a broader view of agility, encapsulated by the following 4 core values, listed in the agile manifesto (Fowler and Highsmith, 2001):

- Value individuals and interactions over processes
- Value working software over documentation
- Value customer collaboration over contract negotiation
- Adapt to change over following a plan

These 4 agile values present inherent obstacles and challenges to decision making with which development teams grapple. While some research outlined below has examined decision making related to these 4 values, there is still much we do not know.

The first value's focus on individuals and interactions means that the ASD team structure is regarded as "organic and flexible" rather than "mechanistic, bureaucratic and formalised". Specialised roles can hinder decision making at the operational level on agile teams (Moe and Aurum, 2008) and so developers are not confined to specialised roles and typically self-organise, interchange and blend roles (Nerur et al., 2005, pg. 75). Decisions can be made outside of members' traditional skill areas and include changing requirements, which may occur daily; problems identified that require a resolution; and new ideas that are generated which need to be explored (Austin and Devin, 2009). But we do not know whether an ASD team's organic structure follows a compensatory or non-compensatory decision process for these or other decisions, meaning we do not know how much information they seek and use to make decisions.

As the second value states, ASD deprioritises documentation in favor of product to spend less time documenting tasks and features to aid in the speedy delivery of working software (Fowler and Highsmith, 2001) with agile teams' reduced documentation associated with a less rational approach to decision making (Zannier and Maurer, 2007). Ironically, more structured design problems have been found to use more rational approaches to decision making, whereas less structured design problems tend to use more naturalistic decision making (Zannier et al., 2007). What we do not know is whether communication methods help the ASD team make decisions when there is less documentation on which to rely.

The third value focuses on collaboration with the customer. The project manager's role as a decision-maker is greatly reduced and more akin to that of a facilitator or coordinator (Alleman, 2002; Lindstrom and Jeffries, 2004; Nerur et al., 2005) who works with the customer and the team. The team makes most decisions, creating a "pluralist decision making environment" (Nerur et al., 2005, pg. 76) due to the diverse backgrounds, attitudes, goals, and cognitive dispositions of the team members, including the customer (Chin, 2004; Cockburn and Highsmith, 2001; Highsmith, 2004). There is less focus on contract negotiation. Yet it remains unclear whether all team members and customer participate in decision making or whether decisions are driven by a few key team members.

The fourth value of adapting to change means that the iterative, incremental nature of ASD with frequent product releases enables teams to adapt and respond quickly (Dybå and Dingsøyr, 2008). To date, existing research has focused on the alignment of decisions over time in ASD and the alignment of decisions on a strategic, tactical and operational level (Moe and Aurum, 2008). However, it remains unclear whether these time pressures necessitate more frequent, short-term and potentially pressurised decision making based on inaccurate or incomplete information for other types of decisions related to planning and on-going development besides the initial design decisions researchers have examined (Zannier et al., 2007; Zannier and Maurer, 2006; Zannier and Maurer, 2007).

Thus, while some have researched decision making related to these 4 values, there is still an absence of an over-arching decision making paradigm in ASD environments (Austin and Devin, 2009; Moe and Aurum, 2008; Zannier et al., 2007; Zannier and Maurer, 2006; Zannier and Maurer, 2007). The inner-workings of these decisions - their component characteristics and challenges posed for their decision process, decision intelligence used, and their potential impact on decision quality - receive little or no attention in ASD practice. These are the focus of this study and are described in more detail below.

2.2 Decision Process

Long before ASD, researchers found that using particular decision strategies is dynamic (Häubl and Trifts, 2000; Johnson and Payne, 1985; Kuo et al., 2004; Moldafsky and Kwon, 1994; Payne, 1976). Further, in scenarios where there are many decision alternatives with a number of variables, decision strategies tend to be either compensatory or non-compensatory (Payne, 1976; Tversky, 1972). A compensatory strategy is based on a knowledge of, and trade-off between, values of the relevant variables to a particular decision. Here, suboptimal information on one or more variables can compensate for preferential values on one or more other variables. A non-compensatory strategy considers variables on a threshold value level and eliminates any decision choice that does not meet that variable's threshold. Non-compensatory strategies are easier but typically of lower quality because they do not examine all available information as the threshold level automatically eliminates possible decision choices whose variable(s) do not make the threshold cut-off. The choice to use a particular strategy depends on many factors, including the decision maker's cognitive ability, behavioural characteristics, experience, and the availability and accuracy of information. The nature of the decision making task also leads to particular strategies: the use of accurate but more effortful strategies arises in tasks where the decision outcome has great importance and decision making time is secondary to outcome (Payne et al., 1988).

Non-compensatory decision strategies reduce the amount of information processed for decisions by using cut-off levels for variables and establishing the most important variables. An example of ASD teams using non-compensatory strategies is not creating contingency plans or examining alternatives to information presented and decisions because of prior team success (McAvoy and Butler, 2009). There is an effort-accuracy trade-off between taking the time to access, examine, and evaluate all relevant intelligence to design possible solutions and make a decision or in the interest of time, consider a minimum amount of available information. Compensatory strategies are more thorough because they compare all relevant information and trade off good and bad aspects for each option (Payne et al., 1988).

However in dynamic ASD contexts, the availability and usage of all relevant information is unlikely. Further, the team context presents opportunity for suboptimal decision processes such as groupthink, groupshift, or a propensity to delay awkward decisions until the future (Briggs and Reinig, 2010; Coyle et al., 2011). It seems likely that ASD decision makers would struggle to make full use of available information, and where relevant information is not immediately at hand, ASD's time criticality may discourage information retrieval. How ASD teams use decision intelligence, and how much they employ, becomes an evident precursor to the strategies used in decision making and is discussed next.

2.3 Decision Intelligence

Decision intelligence describes all valuable sources of input for particular decisions (Coyle et al., 2015; Davern et al., 2008; Shim et al., 2002). In the ASD context this encompasses documented information (e.g. hard- and soft-copy documents, email trails, repository text, code comments, and records of meetings); experiential and tacit knowledge of team members; knowledge of formal variables at play (e.g. unforeseen customer changes to requirements, or code dependencies); and non-contextual attributes (e.g. staff availabilities, or funding limits) (Coyle et al., 2015; McAvoy and Butler, 2009). It also includes verbal traffic shared through team interactions, behaviours, customer feedback, ad hoc conversations, email and instant messaging systems, whiteboards, and project management systems (Coyle et al., 2013). Prior to ASD, Zmud (1979) argued that decision makers do not understand their information requirements and can demand too little or too much information. More than 30 years on, this remains an issue particularly relevant in ASD considering ASD's quick times to market, short development periods, regular software releases, and incessant time (Fitzgerald et al., 2006; Fowler and Highsmith, 2001).

Likewise, there is a double-edged sword: because ASD minimises documentation (Fowler and Highsmith, 2001), how decisions are made is often not recorded, and consequently the quality of that form of intelligence for subsequent similar or related decisions can be adversely affected (Talby et al., 2006). Further, with ASD decision intelligence being multifaceted and largely unstructured, there are clear challenges for quality decision making, which we discuss below.

2.4 Decision Quality

Researchers agree that the decision making process impacts decision quality (Dewan and Hansen, 1994; Häubl and Trifts, 2000; Huang et al., 1999; Montazemi et al., 1996; Sturman et al., 1996; Todd and Benbasat, 1993). Poor decisions in ASD can result in code backlogs, incomplete work, bugs, and inaccurate estimates. The decision strategies employed under ASD methods can directly impact decision quality (McAvoy and Butler, 2009). In ASD, decisions are made on-the-fly, with decision choices made during and between iteration planning meetings (IPM), and at times completely reversed (Drury et al., 2011). Decisions tend to necessitate information on a multitude of potentially informing attributes, but in the ASD context decision intelligence may not easily present access to underlying data. When data on various variables are available, decision makers may trade off information on some variables with other variables, for example the need to complete a code package within 2 days versus the availability of a certain developer. In these instances decision makers apply a compensatory process, typically associated with higher quality decision making (Payne et al., 1988; Todd and Benbasat, 1999). However, where decision intelligence is too difficult to access or where the relevant attributes are partially unknown, non-compensatory decision processes dominate: they take less time because less information is comparatively examined, and 1 or 2 key decision variables outweigh the consideration of others, for example estimating a development task at 2 days because that particular software developer has capacity without information that a parallel project is running late and may be likely to temporarily "grab" that developer during those days. The strategy employed is a powerful indicator of decision quality.

Additionally, many subjective aspects to decision quality exist in team-based decision scenarios. Post-choice decision satisfaction is a common one (Goodhue and Thompson, 1995). Decision satisfaction refers to both satisfaction with the decision process and the decision outcome. It encompasses perspectives of the persons and team building the software and the end user customer. Satisfaction with the decision process is assessed by how extensively a team member participated in the decision

making, how many alternatives they evaluated, and whether they felt rushed in deciding. Satisfaction with the decision outcome is assessed by whether decision outcomes are relevant and useful to both the decision maker and end user, and whether they feel outcomes are reliable and comprehensive (Paul et al., 2004). In ASD, the customer's satisfaction is crucial as it drives development choices in future iterations.

Research on decision quality has not been consistent with either the selection or usage of decision quality indicators (Coyle et al., 2011; Coyle et al., 2013). Some research focuses on more objective indicators such as time and outcome (O'Riordan et al., 2012), whereas others focus more on subjective indicators of satisfaction (Bingi et al., 2001; Dennis et al., 2001; Moreau, 2006). The inconsistency becomes more problematic in the ASD context as the very nature of agile poses disparate behaviours across teams and organisations. It then becomes a challenge to identify an appropriate set of decision quality indicators for the ASD environment.

Thus, considering the 4 agile values and our tripartite decomposition of decision challenges, Figure 1 outlines our framework for the study:

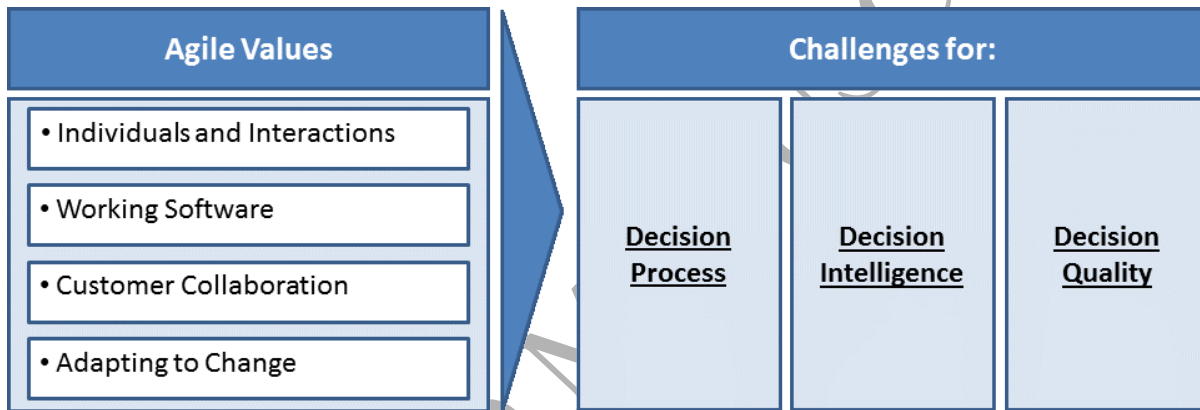


Figure 1. Research Framework

3 METHOD

We conducted an in-depth exploratory case study (Stake, 2000; Yin, 2003) using multiple methods of data collection (Benbasat et al., 1987). This is particularly appropriate in the ASD context where decision making behaviours are complex, dynamic, and highly social.

3.1 Case Selection

The case is a market leader for corporate actions and custody solutions to the investment services industry, employing 300 people, with a client base including global custodians, sovereign wealth funds, and large asset managers. The team studied is responsible for the component that controls the automated generation of messages to notify appropriate stakeholders of various events related to a corporate action, e.g. the creation, updates, and approaching dates of corporate action. The case was purposefully selected for a number of reasons. First, the team was applying agile in a very complex, challenging environment where high profile clients with high demands were implementing the developed software in live financial transaction environments. Therefore, decision making was critically important with high accountability and repercussions if decisions were sub-optimal. Second, the team allowed ongoing, in-depth access to staff through interviews, observations, and document analysis. This proved highly valuable when studying agile decision making which is often subtle,

nuanced, and deeply embedded in social interactions. Third, the team studied had used Scrum for 2 years. The length of time a method is in use is a potential threat to validity. However an analysis of agile method studies reveals that most are using the method in question for between 3 months and 3 years, and so we regarded 2 years of use as reasonably mature. Also, given that staff turnover is high in software development, it is unusual to find a case such as this where all team members had been members for this period.

The team was distributed, adding to the richness of the decision making environment, with 6 members working in their Dublin office, 2 working virtually from their homes in Ireland, and 1 working in India. The team members included 1 scrum master (SM), 4 software developers (SD), 3 quality assurance (QA) testers and 1 business analyst (BA) who represented the customer and their needs (see Table 1). Software development iterations occurred in 2-week iterations involving the typical Scrum practices of product backlog, sprint planning, Scrum board, collective code ownership, daily stand up meetings, and sprint retrospectives. They used Jira, a common bug and project management tracking software tool.

Case Attributes	Description
Team composition	Years of Software Development Experience 8 members in Ireland: 1 scrum master (SM) 12 4 software development engineers (SD): - SD1 10 - SD2 4.5 - SD3 4.5 - SD4 5 2 quality assurance testers (QA): - QA1 13 - QA3 8 1 business analyst (BA) 8 1 member in India: 1 quality assurance tester (QA2) 6 Average Experience: 7.89 Median Experience: 8
Project duration	10 months (comprised of 2 week iterations)
Type of system	A corporate actions automation solution that streamlines operations flow and aids decision making when intervention is necessary
Customer	External, large financial intuitions handling corporate action events
End users	Individuals involved in making and communicating corporate action events at large financial institutions

Table 1. Case Study Profile

3.2 Data Collection

Data collection took place over a 6-month period involving semi-structured interviews, team observations, and project document and photograph collection. Figure 2 illustrates our case selection and data collection processes with Table 2 providing the length of time for interviews and observations. Interviews included 18 individual, face-to-face, semi-structured interviews (2 per team member) and 2 focus groups (with the majority of the team present). The questions were open-ended, allowing respondents to convey their experiences and views of the socially complex contexts that

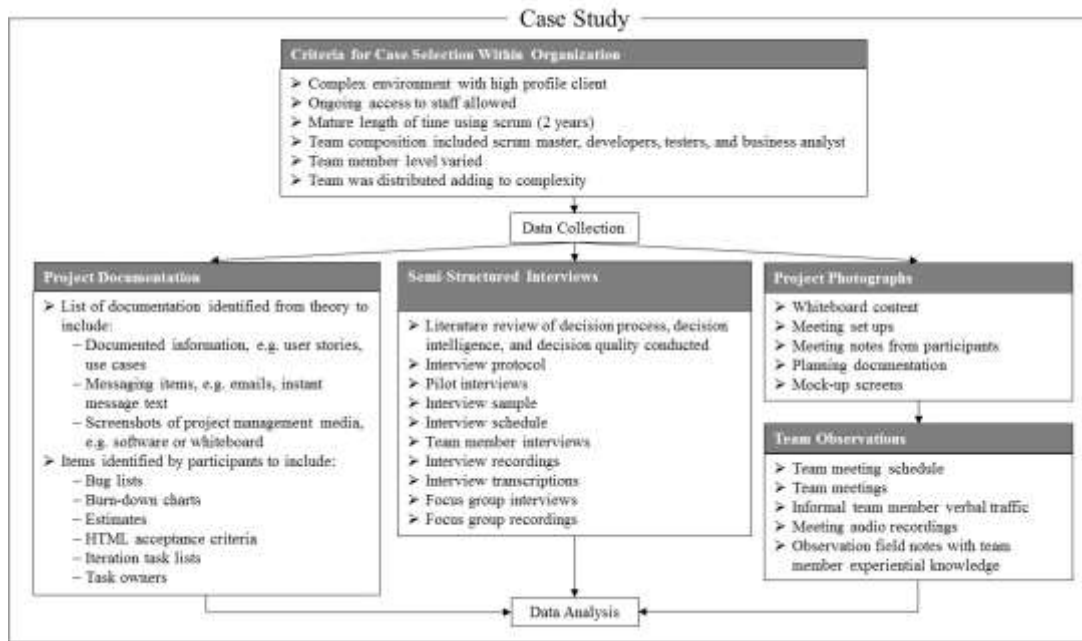


Figure 2. Case Study Methodology

Data Collection Method		Description		
Semi-structured interviews		Duration in Hours		
		Min	Max	Mean
	• 18 (2 per team member)	0.75	1.33	1.10
	• 2 Focus Group sessions			
	– Focus Group A	-	-	0.50
	– Focus Group B	-	-	0.33
	Focus Group Composition	Years of Software Development Experience		
		Min	Max	Mean
	• Focus Group A (6 participants): SM, SD1, SD2, SD3, SD4, QA1	4.5	13	8.17
	• Focus Group B (7 participants): SM, SD1, SD2, SD3, QA1, QA3, BA	4.5	13	8.57
Team observations	Team Meetings	Duration in Hours		
		Min	Max	Mean
	• 3 Story Elaboration Meetings	1.00	3.75	2.60
	• 4 Daily Scrum Meetings	0.25	0.25	0.25
	• 5 Retrospective Meetings	0.25	0.50	0.35
	• 9 Iteration Planning Meetings	0.50	3.50	1.75

Table 2. Data Collection Profile: Time Length of Interviews and Observations

underpin ASD (Oppenheim, 1992; Yin, 2003). The interviews were conducted in a responsive manner (Rubin and Rubin, 2005; Wengraf, 2001), allowing the researchers to follow up on insights mid-interview and adjust content and scheduling of interviews accordingly (see Appendices A-B for sample protocols used in the interviews and focus groups). The first round of interviews took place before an IPM where many decisions were made, whereas the second round of interviews took place after an IPM and after data coding had begun to clarify and confirm data. This was intended to capture valuable data about the purported and experienced decision process. The 2 focus groups were conducted after 2 IPMs to capture feedback about the decision process the team members had immediately experienced. All interviews were recorded with interviewees' consent and subsequently transcribed, proof-read, and annotated by the researchers. Additionally, we observed 21 team meetings on a fortnightly basis, and collected team member experiential knowledge and verbal traffic shared through team interactions during observed meetings. We also collected supplementary documentation to support the interview data and took project photographs to provide additional data for analysis (see Figure 2 for items collected).

3.3 Analysis

Data analysis was based on Strauss & Corbin's (1998) open coding and axial coding techniques. Figure 3 illustrates this data analysis process and Appendix C provides a research audit trail to support this methodology. The interview questions and subsequent analysis were based upon Payne's (1988) compensatory versus non-compensatory decision making model applied in the context of the 4 agile values (Fowler and Highsmith, 2001). These agile values provided an initial list of "intellectual bins" or "seed categories" (Miles and Huberman, 1999) to structure data collection and the open coding stage of data analysis.

Through the lens of these 4 agile values, the 31 individual interview questions, 10 focus group questions, project documentation, and photographs provided further categories for the ASD decision characteristics studied. NVivo was used to store and analyse the data. Decision characteristics were identified from a number of sources: some were explicitly stated by team members whereas others emerged from the interview data and observations. When there was disagreement between observations and comments, this was noted in the findings. At this stage, we identified 13 open codes in which we coded all of the data (see Appendices C-D for coding examples).

The second phase used Strauss & Corbin's axial coding (1998). As the data were coded, theoretical questions, hypotheses, and code summaries arose. These were documented in analytic memos (Miles and Huberman, 1999) to aid understanding of the concepts studied and refine further data collection. This step of the coding process highlighted the 3 categories of decision challenges in our final framework: decision process, the decision intelligence used in decision making, and decision quality.

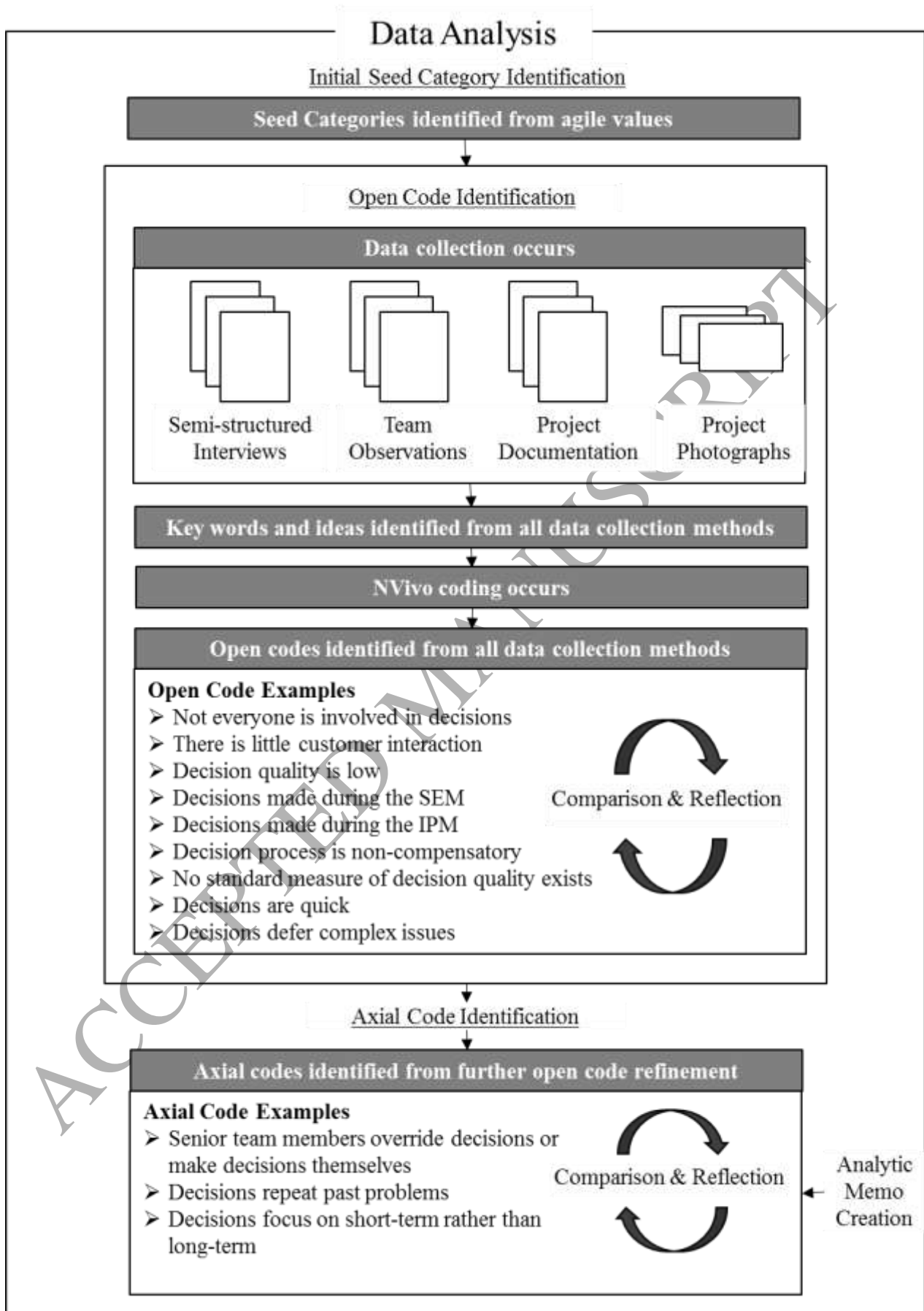


Figure 3. Data Analysis Process

From here we used the 13 open codes from Round 1 to further refine the decision characteristics driven by agile with details of these characteristics under each of the 3 categories of decision challenges (see Appendices C-D for audit trail and coding examples). As categories emerged, the second round of 9 interviews, 1 per team member, elicited further, richer, more focused information. This confirmed, extended, and sharpened the evolving list of categories. This stage of data analysis led to the 12 variables agile teams should consider when deciding to take on tasks during IPMs (see Figure 4). As categories became integrated, further data collection did not identify any additional categories and were deemed to be theoretically saturated (Strauss and Corbin, 1998), and data collection ended. By this point, we determined a final 7 agile decision characteristics under the 4 agile values that formed the leftmost column in our framework. It also led to the sub-categories of these decision characteristics under each of the 3 decision challenges. In all, the case study database fulfilled recommendations by Yin (2003) with its components of case study notes, case study documents, tabular material, and a case study narrative.

3.4 Threats to Validity

In order to evaluate the validity of this study, the authors have used the validity framework presented by Wohlin (2012). Construct validity relates to obtaining the right measures of the concept being studied. The main threat in this context was that the researcher influenced the outcome of the study with his or her presence. While none of the authors were part of the agile team studied, the authors' presence during data collection, e.g. conducting interviews and observations, could have influenced the team members. This threat was reduced by data triangulation, providing multiple measures of the same phenomenon (Yin, 2003). The concepts and terminology of decision making were also reviewed and explained to interviewees for 30 minutes prior to interviews to ensure interviewees were familiar with these concepts. Conclusion validity relates to the ability to draw the correct conclusion between theory

and observation. This was reduced by (i) involving 2 researchers in the interpretation of interview data, (ii) maintaining a full "audit trail" from data to conclusion, and (iii) reviewing the findings with a senior agile practitioner. Also, the interviewer allowed participants to convey their opinion but ensured that each opinion had some evidence drawn from the project being studied.

External validity relates to the extent to which results are generalisable. Generalisability is rarely a goal of case research where there is usually a rich and complex context. While we did not intend to generalise and our method does not support generalisability, some findings are likely to be applicable to other teams. Likewise, to aid at least some generalisability, it is extremely important to describe the features of the research context (Petersen and Wohlin, 2009). Table 1 giving our case profile describes the most relevant subset of elements for the conclusions drawn in relation to the objective of this study.

In terms of internal validity, the fact that data was collected at different points in time is one potential issue. To address this to the extent possible, a number of recommendations were adopted from Miles & Huberman (1999). The interview protocols specifically included temporal questions to ensure data collection at each point considered any changes in context since previous collection points. Also, the authors reflected after each point of collection to ensure data collection instruments were continually updated to reflect the changing context. Finally, findings from previous data collection points were periodically raised with interviewees to reassess whether the findings were valid.

4 RESULTS

Seven key characteristics of decision making in ASD emerged in relation to each of the 4 agile values. We also examined the corresponding decision challenges for each decision characteristic (see Table 3). Italicised text in this section delineates the bullets summarised in the table.

4.1 Value Individuals and Interactions Over Process

Fowler and Highsmith's agile manifesto (2001) fosters a belief in the importance of interactions between individuals rather than a specific process. In this study this value was evident in that decision making was left to the dynamics of individual personalities and interactions rather than any defined prescribed process. This freedom resulted in the emergence of 2 key characteristics of decision making, namely (i) the repetition of decisions during and across iterations and (ii) domination of more experienced staff in decision making activities. Each is now discussed.

4.1.1 Decisions repeat past problems

Many decisions were often repeated, with no deliberation about whether decisions should be different, thereby repeating prior problems. For example, the decision to continually allocate the same task to the same person without discussion was common. Experience was continually a key variable considered when allocating tasks, with those selected often possessing "good experience in that particular area" (SD4). There were 45 references to this fact across 7 of 8 interviewees. Someone decides to take the task if they have worked on a similar task in the past: "I did something like this before, I know how to do it...I can do this quick[ly]" (SM). This was a problem for some who stated that they wanted to take a task to "gain experience" but were often "under pressure to get stuff done" (SD1) so took a more familiar task instead.

This was also observed in the 9 IPM meetings and our review of project documentation and photographs. From our IPM observation notes and photographs, we observed that in most instances tasks were automatically reallocated to the respective person who worked on that task previously. This was confirmed by the team members, and also evidenced by an analysis of the task list and task owners for prior iterations on Jira, the project management tracking software tool. As such, a *non-compensatory decision process is used* and the *decision process repeats prior similar decisions*. This poses a potential challenge in that while it may be the simplest and easiest process for the team, they do not consider all information or alternatives for a decision, and team members do not learn new functionality because they do not take on new roles. The BA admitted that "ideally you'd like to spread the knowledge across the team" so everybody increases skills, but the team struggled with this balance of whether someone had "worked on this area before...". Through this research, we identified 12 variables that team members should consider when deciding task estimates and owners (see Figure 4).

When decisions were repeated, challenges existed because (i) *no standard measure of decision quality exists* for this team and therefore (ii) the *quality measure is not visible to the team*. Based on observation of 3 story elaboration meetings, 4 daily scrums, 5 retrospective meetings and 9 IPMs, the only visible metric was the burn-down chart indicating time left versus work to do. All interviewees indicated there was an inconsistent process for measuring quality. According to all interviewees, the quality of decisions was based on an "unclear mix of customer and team satisfaction" (SD1), "output and output rate" (SM), "defect rate" (SD2), "code quality", or "time to make decisions" (SM, SD1, 2, and 4), although none of these were explicitly measured. Some members believed "decision quality was low when estimates were not being met" (SM).

Decision Characteristics of the Four Agile Values		Challenges for:	
<u>Value Individuals and Interactions Over Process</u>	<u>Decision Process</u>	<u>Decision Intelligence</u>	<u>Decision Quality</u>
Decisions repeat past problems	- Non-compensatory decision process is used - Decision process repeats prior similar decisions	Quality measure is not visible to the team	No standard measure of decision quality exists
Team members with more experience dominate decision making	- Team members are inhibited from deciding to take on tasks or give estimates - Junior members give inaccurate estimates	No relevant challenge identified	Less experienced team members participate less in and are less satisfied with decisions
<u>Value Working Software Over Documentation</u>			
Decisions are made via poor communication and documentation methods	- Decisions are made on incomplete understanding of functionality - Extra work is created for the BA to explain business functionality in lieu of documentation	- No support system exists to automatically present all data with a single, clear, accurate view - Duplicated manual data entry leads to errors tracking task progress	Planning and technical decisions may be wrong due to inconsistency of data
<u>Value Customer Collaboration Over Contract Negotiation</u>			
Level of customer's (or their representative's) experience drives decisions	No relevant challenge identified	No specific customer-oriented data are presented to the team	Anecdotal rather than objective customer satisfaction drives team satisfaction with decisions
Customer interaction is lacking when team only interacts with BA	- BA determines rework and prioritisation decisions - BA provides business knowledge and SM provides technical knowledge for decisions	No relevant challenge identified	No relevant challenge identified
<u>Adapt to Change Over Following a Plan</u>			
Decisions have a short-term focus	Decisions are made for a specific period and then ignored or changed during that time	Decisions made mid-iteration are not tracked or documented	No relevant challenge identified
Quick decisions defer discussion about complex issues	- Some decisions have inadequate diligence - Complex functionality decisions are deferred	Necessary information may not be visible to all team members	No relevant challenge identified

Table 3. Framework of Decision Characteristics and Challenges

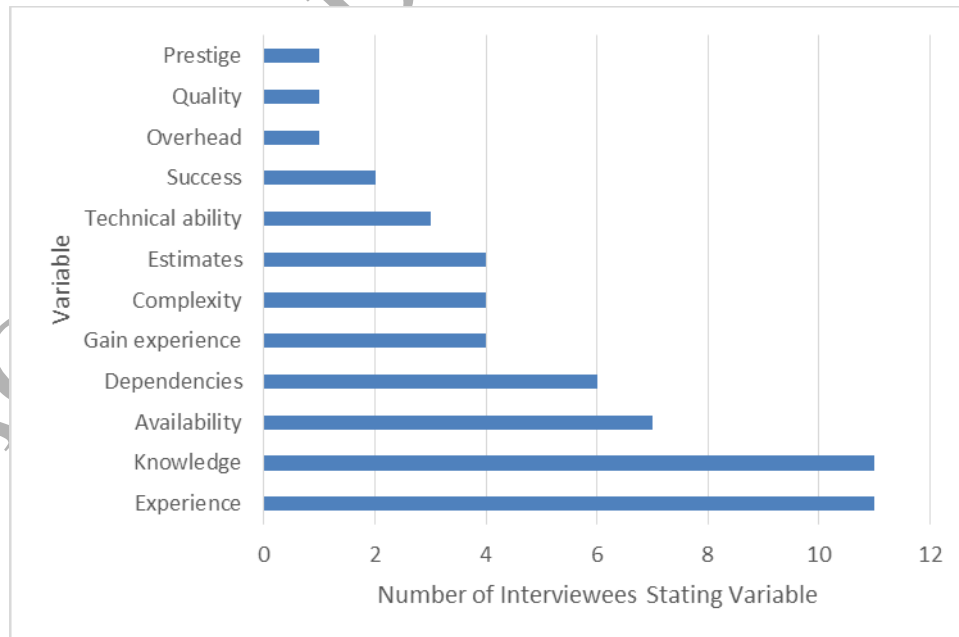


Figure 4. Variables to Consider When Choosing Tasks

Developers stated that their measure of decision quality was limited to the outcome only “if the result is good” (SM), if they “delivered good, working functionality on time, then we made good decisions” (SD4). “Deliver what you say you’ll deliver...on time” (SD1) and if “at the end of the iteration if everybody’s happy and all the functionality is delivered, we know we made good decisions” (SD1). The other evident measure of quality was decision making speed. “Decision quality is high if decisions were made quickly” (SM) because of “efficiency...if decisions just made the whole team more efficient” (QA1) and “[these decisions] save time” (QA1). Alternatively, decisions were considered low quality “if a decision takes up more time, if there’s a lot of time wasted on it... you’re wasting time or else you’re saving time” (QA1). The team defined the quality of their decisions as high when they made decisions quickly.

4.1.2 Team members with more experience dominate decision making

According to the SM, there was a general rule that all team members participated in decision making in this team. But based on observations of 9 IPMs and interview data from 6 of the 8 members, certain senior, experienced team members dominated key decisions, e.g. task allocation, task design, and estimation. Despite the SM’s assertions to the contrary, and in line with groupthink, the team’s general consensus was that the SM “has the say on what has to be done” (SD3) or “suggests somebody take a task” (QA1) rather than allowing junior members to decide to take tasks. The SM decided the iteration’s focus: the “issues to address...the stories to handle...and any bugs to fix” (SD4). The SM “decides the initial task” and “gains consent of the corresponding stakeholder...the developer or QA” (QA2). The BA “weighs heavily” and “drives the product tasks” (QA1). Junior members felt their input was not validated because “when they did raise concerns about complex tasks, it often falls on deaf ears” and senior members “don’t do much about it” (SD3).

Observation of the 9 IPMs and references from senior and junior team members supported this. The senior team members, (e.g. the SM, the senior SD and senior QA tester), decided who should take a task and their estimate for task completion. The SM stated, “I’m trying to complete as many tasks as possible in the iteration to keep us on track with the client’s wants” (SM). Likewise the junior QA member would “defer to me because I’d be the Senior QA” (QA1) rather than make a decision himself. The decision process was challenged as *team members are inhibited from deciding to take on tasks or give estimates*. The SM was the “main person who decides” (SD4). Junior members often took on “smaller tasks” (SD3) and were not involved in decisions about more complex functionality. For example, in 3 Story Elaboration Meetings observed, the SM, BA and senior SD met to decide how to develop new components of the functionality. They then communicated their plan to junior members working on related tasks rather than involving them in the decisions. According to junior team members (SD3, SD4), by not being involved in the meeting, they were unable to contribute ideas or bring up dependencies of which only they were aware from working with that particular piece of functionality.

Regarding decision quality, *less experienced team members participate less in and are less satisfied with decisions*. A junior member said he could not tell senior team members what to do because they would “not respond well to that” (SD2). For example, there was a problem with team members not inputting their tasks in a timely fashion into Jira. Rather than contribute to the decision of how to resolve this issue, SD2 just made sure his own updates were accurate and timely, hoping others would simply follow his example. This indirect way of providing input was not noticed and the problem continued in future iterations.

It was clear from the data that junior staff may not always participate as they have inadequate experience: “Early on I don’t know what we are trying to accomplish because the stories are so complicated, and so I take smaller tasks to facilitate bigger tasks...so I take what tasks the SM assigns me” (SD3). However, not incorporating feedback from other team members at key moments meant critical information was not reviewed for a client release: “If we have a release going to a client, then it [the decision] would come down to myself as to whether the client needs this or whether these issues are critical for them...but the gap [in information from team members] means there’s no context around a decision as to why it was made or if there were errors...that is in people’s heads” (BA).

Observation of 9 IPMs and interview data from 4 interviewees and 1 focus group also indicated that *junior members give inaccurate estimates*. While junior developers could give inaccurate estimates due to inexperience, in this situation, junior developers explained that their decision inputs were not explored or included and senior members changed their estimates. Thus, junior developers saw no point in more effortful consideration of all the variables to decide their estimate because of the likelihood that their suggestion would be overridden. This situation prompted non-compensatory decisions at the individual level, exacerbated by time pressures on team members, a seemingly inevitable backlog, and the belief that decision inaccuracies can be recovered in later sprints. This was an easier but less accurate approach commensurate with non-compensatory decisions strategies and illustrates the snowballing of inaccuracies that in themselves provide an unwarranted veil of legitimacy to final decisions because it could seem as if the senior member considered a number of individual-based estimates to arrive at a considered decision of the final estimate as if using a compensatory decision approach. Whilst the employment of a compensatory strategy is not something explicitly ever in the forefront of the team's approach, the subconscious or conscious belief that a number of options were considered can give a false impression of quality.

Junior members usually decided a default answer of “2 days” for estimates because “knowing what the problem is first of all and then knowing what the code does, you might not have experience with that code, [or know] exactly how to solve it. So that’s why I think a lot of the time, when you say 2 days, you really don’t understand what is involved in dealing with a task, and it gives you time to figure it out” (Focus Group A, SD4). Consistent with groupshift, a senior developer queried the estimate or responded with a revised time estimate for him/her to complete the task based on how long the senior member thought the task should take because “the last thing we want to do is give a short estimate” (SD1). A junior developer admitted giving a “big estimate” (SD3) because the senior developers revised estimates. This junior developer tried to cushion his estimate to give himself time to research and develop his piece of functionality so that the senior members’ revised estimates might still include more time considering the junior member’s skill level. Although short estimates increased the ability to deliver more functionality at the end of the iteration, if they were too short for someone’s experience, functionality was not delivered on time to the customer.

4.2 Value Working Software Over Documentation

Fowler and Highsmith’s agile manifesto (2001) fosters a belief in the importance of minimal documentation. One key related characteristic emerged in this regard: decisions are made on poor information intelligence.

4.2.1 Decisions are made via poor communication and documentation methods

Because the team aimed to minimise documentation, decisions were made using ad hoc, inaccurate,

incomplete, or non-existing documentation. This finding was supported by 32 references across all 9 interviews and a focus group, as well as a review of project documentation and photographs. The decision process was challenged because *decisions are made on incomplete understanding of functionality* as the team used multiple communication channels that were not coordinated. Data were lost and decisions forgotten due to a lack of documentation. Many decisions took place via email or instant message as opposed to the central Jira repository and were rarely tracked or “officially recorded” (QA1). In reviewing the team’s supplemental documentation, we found that documentation for decisions included unclear and inaccessible customer documentation. For example, user stories, the bird’s eye view of how everything fits together, were unclear and there was “a domino effect onto everything else... it’s affecting everything: what you volunteer for, how long it’s going to take, the bugs and then the coordination of the testing” (SD3). They also wanted them “written plainly, in a natural way, not ‘if this, then that’ . . . and there’s very little in terms of context” (Focus Group B, SD2); the “if, then” statements did not give enough detail to understand the use case. The inadequate format of this documentation caused problems as team members “don’t have any faith in it to know exactly what has to be done” (SD4). Acceptance criteria were written in HTML format that was difficult to access as team members had to open multiple HTML pages rather than 1 document. In every iteration prior to the 5 retrospectives we observed, this caused “constant drag on the BAs to say what they need or what they have to do” (BA). *Extra work is created for the BA to explain functionality in lieu of documentation.*

Decision intelligence was challenged as *no support system exists to automatically present all data with a single, clear, accurate view* to co-located and non-co-located team members. The whiteboard was not visible to non-co-located members. Interview data from the SM, developers, and QA showed the team agreed that decisions should ideally be tracked using the Jira tool for all to view. Our observation of 9 IPMs and 4 daily scrum meetings characterised iterations by on-the-fly decisions supported by incomplete data in Jira, emails, and ad-hoc conversations. One noted “we need some sort of repository where everything is all together because task-related problems arise over email and are not inserted into Jira as further bugs, leading to inaccuracies in task completion times and bug counts” (SD3). Jira could act as such a store, but its potential was underused. Often, these decisions and changes were not input into Jira. The team did not have an integrated approach to decision making and tracking facilitated at IPMs. Project metrics were therefore incomplete.

While the whiteboard is a common process in ASD teams, the team noted inconsistencies with the data in Jira and the “updates” captured by the physical Post-Its on the whiteboard as *duplicated manual data entry leads to errors tracking task progress*. This was evident across all interviewees and the project documentation and photographs we collected. Team members were wasting time doing the same tracking in 2 media, and errors and inconsistencies increased as they “did not update both [media] simultaneously and Jira doesn’t truly reflect where we are in a story or iteration” (QA1). The inconsistencies exist because some team members are more disciplined in updating both medium simultaneously. Human error led to inconsistencies as well. Although Jira is more critical to update as it has links to tasks and issues and “Jira is considered the master” (SM), the “whiteboard is a lot more useful because it more clearly lays out what the problem is” (SD2). With the inconsistency of data between text in Jira and the whiteboard, *planning and technical decisions may be wrong due to inconsistency of data*. Without enforcement, Jira tasks and bugs did not contain the most up-to-date and accurate information and were unreliable: “If I’m not properly updating Jira for my task then the decision may be wrong” (SD4). Three team members suggested an interactive whiteboard that integrated with Jira would be a useful tool for IPMs as it may be more accessible from a distributed team perspective because “Jira and the whiteboard always do get out of sync because people won’t

move their tasks and because JIRA is the one that has to be kept up to date, it is generally the board that falls behind. But if we had an interactive whiteboard constantly projected on the screen that would be much preferable” (SM). An “interactive whiteboard would make the planning sessions go a lot quicker” (SD1) and would avoid the “additional task of keeping these 2 [Jira and whiteboard] in sync” (SM).

4.3 Value Customer Collaboration Over Contract Negotiation

For the agile manifesto’s value of customer collaboration (Fowler and Highsmith, 2001), we found the related decision characteristics show decisions depend heavily on the technical experience level of the customer or their representative, e.g. BA, whose team interaction meant less customer collaboration.

4.3.1 Level of customer’s (or their representative’s) experience drives decisions

While the customer might know what they want from a business perspective, this is not always communicated accurately to the team or technically feasible. From 3 interviews, it was clear that the customer often created changes in the requirements for future development, but they were not always clearly communicated to the team, leading to poor decisions and development mistakes. The BA looked for what “the response is from the client” (BA) although the use of this intelligence was challenged when *no specific customer-oriented data are presented to the team*. Although the acceptance criteria existed in HTML format, much of the data from the customer were “in the BA’s head or documented in his notebook” (QA1) and therefore not accessible to the team. Another team member made a similar comment: “these data are in the BA’s head and not necessarily recorded anywhere for us all to see” (SD1). The customer experience and expectations needed to be presented and available to all team members in order to develop and test functionality accurately. This was also observed in the 9 IPMs and 4 Daily Scrum Meetings: customer requests and information only came from comments by the BA; they were not documented for the team to examine themselves.

Decision quality challenges existed as *anecdotal rather than objective customer satisfaction drives team satisfaction with decisions*. The team was specifically asked how they knew if they made good decisions. They believed they made good decisions if both the team and the customer were satisfied with the end product and the process of getting there, which was stated across 17 references from 6 interviewees. But customer satisfaction and team satisfaction were not measured objectively. The only indicator the team had was from the “[general] response from clients, from internal stakeholders, and if there are any remaining gaps in what you’ve delivered. That’s really where you know you’ve made mistakes” (BA). According to the SM, the overall customer satisfaction referred to the “usability” and “performance” (SM) of the functionality. A QA tester stated, “when the client comes back to you and there’s no bugs, no change requests, that’s really when we find out if decisions are successful or not” (QA3). Further, the team simply waited for this customer feedback. They did not seek it out or have a process to receive it after each iteration.

4.3.2 Customer interaction is lacking when team only interacts with BA

The ASD team collaborated with the BA, the customer representative, rather than the actual customer. ASD calls for more collaboration with the customer, and in this case the BA worked very closely with the customer and represented him/her to the team as a direct surrogate. Five interviewees indicated that the team viewed the BA as the “owner of requirements because he’s the one talking with the client and who gets all the tracking requirements down from the client. So he’s the one who makes the decision, ‘Ok it should be done like this.’ Definitely, his decision depends on the consent of the client.

He'll definitely speak to the client and get consent, but for me he's [the BA] the business owner" (QA2). Here the BA *determines rework and prioritisation decisions*. Instead of the customer interacting with the team, the BA does and it was noted that he "knows especially what [the client] needs and what has to be done" (QA3). Furthermore, "if there's any business questions from my point of view, I'd be asking the BA what has to be tested, the priorities, you know, the business value. Then I'd write test cases and these ones have to work for [the client]" (QA3).

Yet the BA and the customer often do not have the technical experience to understand the functionality of requirements, so the SM contributes his opinion of the task complexity and the team's capability to complete customer requests, which one member stated as "the reality of developing" (QA1). Because of his technical knowledge, the SM helped the BA make prioritisation decisions because the BA "knows what's more important for the client" and the SM "knows how to do it technically" and therefore "they tell us [the team] what to do first in the iteration" (SD3). Another developer supported this: "If I'm developing something, I usually ask the BA and we make decisions. If it's fine for him, then it's usually fine. But if he has any doubt, then we ask the scrum master" (SD2). Within the decision process, the BA *provides business knowledge and SM provides technical knowledge for decisions*.

4.4 Adapt to Change Over Following a Plan

Finally, the agile manifesto values responsiveness to change (Fowler and Highsmith, 2001). While this is core to agile thinking, this study found 2 related decision characteristics arising: (i) short term decision focus and (ii) deferred discussion of complex issues.

4.4.1 Decisions have a short-term focus

ASD teams work in short iterations with significant decisions made each iteration. A decision process challenge is that *decisions are made for a specific period and then ignored or changed during that time*. Rather than follow-through with decisions made for an iteration, observation of 9 IPMs and 4 daily scrum meetings indicated that many decisions were re-made mid-iteration, keeping a very short-term focus on decisions. Decisions change quickly during an iteration rather than waiting for the next iteration to change decisions. Interview data from developers and the SM further supported this finding. Mid-iteration, the team may apply corrective action to "change an estimate" (SD4) or if bugs arise from the client, they "completely change the focus of what we are doing [mid-iteration]" (SM). The team tried to "keep this to a minimum" (SD1) to avoid unnecessary rework, but often decisions were made or changed mid-iteration rather than waiting for the next IPM in keeping with ASD's value of adapting to change.

Based on observation of 5 retrospective meetings, a common challenge for decision intelligence was that *decisions made mid-iteration are not tracked or documented*. Team members updated Jira with task and estimation decisions immediately after IPMs but were less likely to do so during the iteration. This was particularly problematic for tasks that "other developers had dependencies on" (SD1). Members continued on tasks that were modified without realizing the modifications: "we'd be halfway through the iteration and realise we should have done this as well" (SM).

4.4.2 Quick decisions defer discussion about complex issues

It was clear from all meeting observations and 14 interviews that decisions were often made quickly as members sought to reach an optimal decision with least effort. Team members admitted they made decisions quickly as the environment was "so frenetic...it's so fast-paced" (QA1). Given ASD time

constraints, interviews indicated that some major decisions only considered a few variables. For example, task allocation was done in a non-compensatory manner, with decisions made on minimal information rather than examining all available information. Regarding decision intelligence, the *necessary information may not be visible to all team members*. For example, when deciding who takes tasks, the SM may be aware of a team member's availability, experience, and technical knowledge, but a lack of awareness of these amongst all team members hindered the agile practice of self-organisation.

There were further challenges for the decision process. One such challenge is that *some decisions have inadequate diligence*. Worryingly, these decisions had the "most risk" or an "external dependency" (SM), for example how to develop complex functionality for notification messages to customers. They held workshop sessions during the iteration to clarify desired functionality with the BA and decide how to develop but felt pressure to decide quickly. However, based on the observation of 5 retrospective meetings, the team often cited the need to give more time to these decisions to get them right the first time because they admitted that *complex functionality decisions are deferred* to future iterations, which is another decision process challenge. They should "spend the extra time researching the requirements and acceptance criteria" in order to have "more faith, more confidence that it will meet the acceptance criteria [now]" (SD1) so it is not "just time wasted" (BA). These findings are commensurate with a non-compensatory approach to immediate decision making, with particular attributes consciously excluded due to time constraints and pressure to advance. Complex decisions are procrastinated.

5 DISCUSSION

The central contribution of this research is a framework for a) understanding the decision characteristics that relate to the 4 core agile values, and b) evaluating the challenges these characteristics present for the decision process, decision intelligence, and decision quality. This builds upon prior research indicating the general decision obstacles that ASD teams face: (1) an unwillingness to commit to decisions; (2) conflicting priorities; (3) unstable resource availability; and (4) a lack of implementation, ownership and empowerment (Drury et al., 2012). To answer both research objectives regarding the particular characteristics and challenges of an ASD team's decisions, the decision characteristics and challenges were identified in the findings section, and are depicted in the framework (see Table 3).

While the findings section was organised to examine the decision characteristics of each of the agile values, we focus the structure of our discussion on the implications for 3 decision challenges, identified as challenges for the decision process, decision intelligence, and decision quality (see Table 3).

5.1 Decision Process Implications

This case study identifies decision process challenges for each of the 4 agile values: (i) better strategies for better quality decisions challenges in team-based decision making processes, (ii) improved documentation for decision tracking, (iii) ways to effect better customer collaboration, and (iv) recommendations to consider the long-term effects of decisions beyond the iteration. Each is now discussed in more detail.

Regarding the values of individuals, interactions, and customer collaboration, we had expected to see individuals collaborating or at the very least sharing the decision making process with all team members. However, experience was identified as a main factor for making decisions as senior

members on the ASD team dominated decisions, often to save time. This facilitates groupthink (Briggs and Reinig, 2010) as one member makes the decision without consulting other team members and their views. Further, as shown in the results section, in instances where team members proffered suggestions, there was evidence to suggest these were disregarded. Decisions about whether an issue was addressed and a story or a bug were done were deferred to the most senior QA team member. All team members were not involved in the decision process, and team members with more experience guided decisions for tasks and estimates. Although research defines the SM as a decision making facilitator between all team members rather than the actual decision maker (Augustine, 2005), our results found the SM still made decisions rather than facilitating team decision making. For example, the SM decided tasks and estimates for team members based on their experience and technical knowledge. This further confirmed related research that found senior ASD team members made decisions without incorporating input from junior members (Drury-Grogan and O'Dwyer, 2013). To facilitate a more collaborative process, this study suggests that ASD teams consider the additional variables for decisions on tasks and estimations, as outlined in Figure 4.

Regarding strategies for decision making, compensatory strategies, although typically more effortful, are superior to non-compensatory approaches, yet the ASD team's approach was non-compensatory as defined by Payne, Bettman and Johnson (1988). This effort-reducing and time-spurred trade-off precluded the team from exploring other avenues with other team members taking tasks whom may have had less experience but could use the opportunity to improve their skill-set with these tasks. With the fast-paced nature of agile, prior research has suggested that an ASD team should maintain a self-organised, flexible team structure (Cockburn and Highsmith, 2001; Nerur et al., 2005), yet McAvoy and Butler (2009) found ASD teams do not examine alternatives or develop contingency plans. Additionally, this research found that an ASD team struggles to remain flexible and examine alternatives in the task allocation and estimation decisions as the SM would decide who would take which task during an iteration. Further, decisions were often repeated. For example, during an IPM the same people were assigned similar tasks repeatedly. These findings are in line with typical team-based phenomena such as groupthink and the Abilene paradox, resulting in team members carrying out tasks not by explicit choice, but because of an implied decision having been made by the team, which in fact resulted from a procedural memory artifact (Coyle et al., 2011). Fundamentally, non-compensatory detail-light decision making is inferior, and the ASD team relied on a false sense of security of being able to apply corrective action for short-term decision making within sprints, but allowing a ground-hog day scenario involving the repetition of poor decisions. Considering that the indicators for quality were minimal and passively anecdotal (section 4.1), there was little to indicate to the team that this approach was inferior, or that superior strategies existed.

Further, 2 key findings emerged regarding the agile values of less documentation: (i) team members make decisions on incomplete information, and (ii) the BA has extra work explaining functionality in lieu of documentation. Both points give further support for a dominance of the non-compensatory decision model. While agile teams believe the cost of creating and updating documents outweighs the benefits of documenting the system in detail when frequent changes in the requirements and source code occur (Chau et al., 2003), the existing customer documentation was poor, and decisions often were neither tracked nor communicated. Off-site team members often did not participate in decisions. Similar to Zmud's (1979) argument that decision makers often misunderstand their information requirements and demand too little or too much information, we found this ASD team did not examine enough information to make high quality decisions. For example, they reduced the options for the team member who would take a task by only examining members' experience and knowledge of the functionality. They did not regularly consider the complexity of the product or gain experience

with the functionality. This study suggests that the application of compensatory approaches to team decision making, cognisant of the inherent need for finer granularity and definition across multiple variables, can reduce the need to revisit decisions later in the software development timeline, and lead to a more robust product through increased ongoing decision quality. Teams should not ‘gloss over’ unknown variable values as fixable later without incurring an indeterminate time and effort for redress later in development. Teams should raise their awareness of the value to compensatory approaches to decision making, which in turn highlight the importance of maintaining ready access to values of the variables at play for a particular decision and clear agreement on what constitutes the quality of decisions. This suggests tracking decisions and the variables’ value, which suggests a better usage of documentation.

Regarding the value of customer collaboration, this study shows that (i) the customer’s experience, or that of their representative (e.g. BA), drives decisions, and (ii) the customer collaboration is lacking when the BA is the liaison with the team. This meant that the BA prioritised work and provided the business knowledge rather than the customer.

Regarding the value of adapting to change, the team made complex decisions quickly and only focused on the short term because their environment is so fast-paced and structured with a series of short, 2-week iterations. As a result, they focused on tactical decisions and ignored strategic decisions. This has implications for the long-term success of the ASD team and its project. While ASD promotes short-term delivery of functionality, the study suggests that ASD teams consider the long-term effects of decisions and look beyond the iteration via strategic decisions to enable project success.

5.2 Decision Intelligence Implications

This study identifies (i) challenges for the usage of decision intelligence, and (ii) ways to better adapt to change in the context of decision making in the agile culture to avoid knowledge gaps. Each is discussed in more detail below.

The results identified challenges for decision intelligence across all 4 of the agile values. Regarding the 2 agile values of individuals and less documentation, individuals on this ASD team had no long-term view of quality metrics, experience, or decisions. This makes sense for an ASD team as they focus on 2-week iterations. Yet iterative development reduced the team’s ability to see consequences of decisions over time. Sources of decision intelligence included soft channels and only some formal documentation: (i) the project management screen output of relevant information to decision makers; (ii) the underlying structure of team and project data to enable and facilitate querying, aggregating, and charting of decision intelligence; (iii) the presentation of project data such as functionality details, tasks, owners, and bugs available to ASD team members when needed; (iv) tacit verbal interactions; and (v) communications traffic concerning project issues. Such diversity and divergence in decision intelligence presents difficulty for decision making when decision intelligence is needed. Considering the need for timely decisions, the frequency of decisions in these environments with short iterations and the team-based approach to decision making in ASD projects according to Fowler and Highsmith (2001), data used by decision makers in agile scenarios needed to be accessible to many people with varying roles and foci to enable team interaction and clear understanding of functionality to facilitate the decision making process.

Considering the challenges for decision intelligence outlined in our results section, a potential solution would be the integration of all available data into one cohesive system. Almost all team members stated the need for accessibility to all relevant data in a way that was usable in order to make

decisions. As the team was geographically dispersed in Ireland and India, the team eventually stopped using the whiteboard and solely relied on Jira as their data presentation and tracking tool, which reduced overhead time from duplicated manual entry of tasks and increased visibility for all team members to tasks, bugs, and progress. But some information was still not presented in a clear, easily accessible manner, namely the ineffective client documentation format that outlined the functionality to be developed. This was a challenge for the customer collaboration value. In terms of explicit documentation, the current HTML format was cumbersome for the team, and they rarely read all of the documentation entirely so they often did not fully understand what needed to be done. They therefore leaned on the BA or developed functionality without understanding the requirements. This had a domino effect on all other decisions, including deciding to take on a task, giving the estimation, and deciding how to develop something. Thus, it was a major issue as it often delayed iterations.

Regarding the agile value of adapting to change, findings show the ASD team struggling to adapt to change as there was no mechanism for tracking decisions. Although Talby (2006) suggests project metrics such as product size to track decisions, our study shows a need to track decisions and their changes beyond existing project metrics because the ASD team could not see the impact of repeating the same decisions. For example, knowledge gaps were created because the senior team members continued to task the same people complete similar functionality. This seemed positive as it helped complete tasks more quickly. Although the team recognised the importance of allowing members to take on a variety of tasks, they did not do this, thereby repeating the same decision. Likewise, when decisions were made mid-iteration rather than waiting for the next iteration, the team did not track changes to decisions and whether decisions improved over time. According to Gale (2012), tracking information becomes more important as the agile project progresses, yet we did not see our ASD team tracking decisions at any stage in the project. Thus another knowledge gap was created because the team had no idea what decisions were made mid-iteration over time or the consequences of such decisions. Although making decisions mid-iteration allows an ASD team to respond to change quickly, these decisions must be tracked properly in order to recall what happened and better define tasks for future iterations.

5.3 Decision Quality Implications

This study identifies challenges for decision quality, and suggests a deterministic approach to metric development for agile sprints. Decision quality challenges existed for 3 agile values (there were none identified for the value of adapting to change) as no standard measure of decision quality existed. This ASD team determined decision quality in an ad-hoc output-focused manner based on whether the software was “good”, the client and team were satisfied, and tasks were completed on time. These measures of decision quality are present in the general decision literature: good output (Todd and Benbasat, 1999), client and team satisfaction (Goodhue and Thompson, 1995; Paul et al., 2004), and timely completion of decisions (Kuo et al., 2004), but they have not been applied to ASD. These findings are commensurate with an outcome-based view of quality where a short *time* symbolises *good* decisions but is in line with non-compensatory decision making. This ASD team determined decision quality via active participation in decisions, as less experienced team members and those who could not view information easily did not participate in decision making. Prior research states that ASD teams are satisfied when all members are involved in decision making, and can own the functionality they work on and its implementation (Drury-Grogan, 2014), and this team agreed that should happen but in practice it did not. As such, there was evidence that poor involvement, disregarded suggestions, inaccurate estimates, and at times groupthink persisted across time to future decisions, providing a seed of suboptimal process as functionality increased over successive sprints.

Further, there were no standard or robust quantitative measures of quality, which relied more on anecdotes. While quantitative data on decision satisfaction were not collected, future research could explore these parameters. There were no in-sprint metrics of satisfaction and progress, and customer feedback came after functionality was completed. We suggest ritualistic embedding of decision satisfaction metrics for development sprints. Considering the impact of poor decision making outlined in our findings, the study also suggests the inclusion of these metrics in retrospectives to provide a cyclic feedback mechanism absent in current development. Further, embedding the customer in sprint activity was absent. It is apparent that a more cyclically-inclusive and robust consideration of customer involvement at all stages of development may reduce corrective action and increase satisfaction.

6 CONCLUSION

There are many ASD characteristics that affect decision making, but there is no holistic, structured manner depicted via an over-arching set of ASD characteristics that affect decision making. This study provides this framework, identifying 7 decision characteristics across the 4 agile values. Corresponding to these characteristics we have also defined the challenges associated with the decision process, decision intelligence and decision quality, thereby addressing our 2 research objectives.

Regarding theoretical implications of this research, these findings extend our contribution to the agile literature. Of our decision characteristics identified, 5 are not necessarily new to decision making, but they are new to decision making on ASD teams and have previously unexamined decision challenges as we outline below. Furthermore, while this study confirmed the existence of previously identified decision characteristics, 2 were newly identified, namely (i) level of customer's (or their representative's) experience drives decisions; and (ii) decisions have a short-term focus. However there a number of challenges which have not been identified previously, namely (i) non-compensatory decision process is used; (ii) decision process repeats prior similar decisions; (iii) quality measure is not visible to the team; (iv) junior members give inaccurate estimates; (v) decisions are made on incomplete understanding of functionality; (vi) extra work is created for the BA to explain business functionality in lieu of documentation; (vii) no support system exists to automatically present all data with a single, clear, accurate view; (viii) duplicated manual data entry leads to errors tracking task progress; (ix) planning and technical decisions may be wrong due to inconsistency of data; (x) anecdotal rather than objective customer satisfaction drives team satisfaction with decisions; (xi) the BA determines rework and prioritisation decisions; (xii) decisions are made for a specific period and then ignored or changed during that time; (xiii) decisions made mid-iteration are not tracked or documented; (xiv) complex functionality are deferred; and (xv) necessary information may not be visible to all team members.

This study's contribution identifies and posits recommendations for these new challenges. Further, the provision of an over-arching framework (see Table 3) allows researchers to further study ASD decision challenges in a structured and holistic manner. Previous studies had tended to focus on individual challenges only.

The study contributes to decision making in identifying the phenotypical attributes in ASD environments conducive to the dominance of inferior non-compensatory decision strategies. ASD decisions are spurred by the need to be fast, and the team has short development cycles suggesting that inaccuracies in decisions can be recovered without substantial impact. Optimal decision making strategies are further hindered at best, and impossible at worst given the (i) lack of documenting how

decisions occur and a resultant tendency to repeat poor decisions, (ii) unavailability of a coherent set of decision intelligence, (iii) reliance upon decision trend and experiences of the few, and (iv) an almost complete absence of robust indicators for decision quality.

There are also a number of practical contributions. This study provides a framework for practitioners to evaluate the decision challenges in ASD environments and reflect on how ASD teams can effectively measure the quality of decisions. This framework can be applied in an evaluative manner, allowing ASD teams to determine the current state of their decision making and what challenges and problems are occurring. This in itself helps to build awareness of the salient issues amongst practitioners. Issues regarding dominant individuals, seniority of individuals, methods of communication, decision making processes, implicit decisions, and an over-emphasis on short term decision making were examples of findings identified in this study that practitioners may take into account when evaluating or improving their teams' decision making activities.

The study also identifies opportunities for future research. Firstly, as Eisenhardt (1989) suggests, theory building from case studies is iterative, and it is only through repeating the process that one can reach "theoretical saturation". Therefore future research could repeat this piece of research through other cases, identifying further factors until this saturation point is reached. Further, upon examination of Eisenhardt's process for such theory building (Eisenhardt, 1989, pg. 533), another extension of this research would be to elaborate this list by deriving and testing measurable hypotheses, thus "sharpening the construct definition and the validity and measurability of the factors".

Further, while the objective of this research was to identify the relevant characteristics and associated challenges of decision making in ASD, further research could adopt a multi-disciplinary perspective to examine some of the underlying reasons for why people behave the way they do in such a context. Concepts such as groupthink emerged and could be further researched. Agile team interactions could possibly be explained by some fundamental psychological, social, and cognitive theories which to date are lacking in ASD research.

In terms of limitations of the study, the first is that this case study is based on 1 team, and so the results should not be generalised to every agile method. Second, as stated in Section 2, teams always adapt a method to their situation, and this inhibits the generalisability of the study even further. Third, it should not be assumed that even if these challenges exist that they are in fact impactful. For example, contemporary thinking on wisdom of crowds suggests that individuals' thinking and acting as individuals may actually reach better outcomes. Therefore, before taking any corrective action, it is important to determine whether the challenges are indeed having actual or potential negative effects on decision making. Finally, regarding the evaluation of decisions, it is challenging to evaluate decisions as in reality, we cannot replicate the exact decision conditions from one case to another, but that is a part of studying working teams in situ in their organisational setting.

This team was also using Scrum for 2 years, and so studies of ASD teams with more or less experience may yield different results. Our review of documentation also was dependent on the team members sharing documentation with us for review. The findings presented are based solely on data gathered over a fixed 6-month period, and therefore suffer from the usual shortcomings of a study that captures a snapshot in time of a story that is much longer. Thus, longitudinal data collection could yield interesting results, particularly in relation to the issues regarding short and long term decision making and decision impact. Future research should seek to develop and apply solutions to resolve the decision challenges identified in this case study. Also, given the exploratory nature of this study and that it was based on a single, albeit revelatory case, future researchers may seek to conduct further

explanatory research into the nature of these challenges and their impact on decision quality and, ultimately, project performance.

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ACCEPTED MANUSCRIPT

9 APPENDICES

APPENDIX A

Sample Interview Questions

Company Interview Guide

Read instructions: *The purpose of this interview is to understand what group decisions are made on agile teams and how data are presented to teams. Thinking about the agile team on which you currently work and your current role, please answer the following questions. First, there are a few demographics questions to answer, and once you've completed that, we will begin the interview. All of your information will remain confidential at all times.*

Start recording.

Demographic Questions

1. Do you develop in iterations on this team?
2. What is the length of each iteration?
3. What is the total duration of this project? (i.e. years, months)
4. How long has this project been underway? (i.e. years, months)
5. How many years and months of experience do you have in the software industry?
6. How many years and months of experience do you have working on an agile team?
7. How many people are on your agile team?
8. What is your current role on your agile team?
9. Is your agile team co-located or distributed? If your team is distributed, list the locations where teams are located.
10. What agile method(s) does your team use?
11. Please give a brief what your team does.
12. Please give a description of the product your team develops.

Interview Questions

Now we will move to the actual interview. We will begin with a few general questions before moving to the more specific questions on how your team makes decisions.

Decisions Made

1. Describe what happens during your iteration planning meetings (IPMs)? What decisions do you make there?
2. What other meetings do you have where you make decisions? Describe specific examples of decisions made in each of these meetings?
3. What are the informal decisions you make (e.g. at the water cooler, not during a formal meeting)?
4. Who makes the decisions on your team?
5. How does your team track decisions over time? How could your team's decision tracking be improved?
6. Of all these decisions you've mentioned, what decisions are you willing to spend more time on going through all information rather than taking less time and going through less information?

Decision Process

1. Describe how your agile team makes decisions?
2. How does one's skill and experience affect decision making?
3. How long does it take to make decisions?
4. How do you compare possible courses of action before deciding on one?
5. Do you randomly select a course of action without looking at any or all information? For what decisions do you do this?

Decision Quality

1. Give examples of good decisions and bad decisions that your agile team has made.
2. How did you know they were good or bad decisions / successful or unsuccessful decisions?
3. How do you measure (evaluate) the effectiveness of decisions?
4. How could you improve your measure of effectiveness of decisions?
5. What obstacles or issues prevent your team from making decisions?
6. How do you overcome these obstacles?

Decision Intelligence

1. What are examples of data / information used to support decisions?
2. How is this information presented?
3. How could this information be better presented?
4. How does the way that data are presented impact the group's decision making?
5. How do you capture and present project metrics?

Stop recording.

APPENDIX B**Company Focus Group Guide**

Read instructions: *The purpose of this focus group is to understand the decisions your agile team makes during an iteration. You may discuss decisions made individually or as a group. Thinking about the agile team on which you currently work and your current role, please answer the following questions. All of your information will remain confidential at all times.*

Start recording.

Focus Group Questions

1. Describe the decisions you make during an iteration. Include specific examples of decisions and whether these decisions take place during specific meetings.
2. Who makes each of these types of decisions?
3. How does your agile team make these decisions?
4. How could your decision making process be improved?
5. How does your agile team track these decisions?
6. How could your agile team's decision tracking be improved?
7. What decisions are escalated?
8. To whom are they escalated?
9. Is there anything else your agile team could do to improve your decision making process?
10. Is there anything else your agile team could do to improve the quality of your decisions?

Stop recording.

APPENDIX C

Research Audit Trail

Task	Description
Literature review	Detailed literature review conducted as part of research process without preconceptions about what to find and without being desensitised by borrowed concepts. This review included the following topics: agile software development, agile values, agile methods, decision processes (including compensatory and non-compensatory processes), decision strategies, decision making method, information used for decisions, decision information presentation, decision intelligence, decision quality, decision metrics, qualitative research (including case study, semi-structured interviews, focus groups, data observation, document analysis, open and axial coding, and validity). In total, we include 63 references in the final manuscript.
Conceptual/theoretical framework	Four concepts emerged as prominent items in our initial research: • Agile software development values • Decision processes • Decision intelligence • Decision quality Namely, our literature review led us to question the decision characteristics related to each agile value and the related challenges for the decision process, intelligence, and quality aspects. This research used a grounded theory approach of study to develop the conceptual framework from the 4 prominent items in the literature as listed above.
Interview protocol and instrument development	We researched literature on semi-structured interviews and developed an interview protocol focusing on the 4 prominent concepts listed above. Interviews were all audio-recorded and transcribed. They were conducted in a responsive manner so the authors could follow up on insights mid-interview. The authors also sought clarifying information within 2 weeks after reviewing the interview transcripts and observation notes.
Participant selection	The case was selected for a number of reasons: • Complex environment with high profile client • Team allowed ongoing, in-depth access to staff for data collection • Mature length of time using scrum (2 years) • Team composition included scrum master, developers, testers, and business analyst • Team member level varied • Team was distributed adding to complexity
Data collection and storage	Data was collected via in-depth, semi-structured, face-to-face interviews (18 interviewees in total, 2 per participant); focus groups (2 conducted with on-site team members); meeting observations (21 meetings observed), and document analysis of supplemental team documentation and photographs.
Raw data	Raw data includes interview audio files and transcriptions, focus group audio files, field notes, site documents (e.g. email messages, instant messages, HTML documents with acceptance criteria, screenshots of project management tool with iteration tasks, screenshots of burn down charts, photographs of various project items such as whiteboard content).
Partially processed data	Coded interview transcripts and key participant responses and quotes; authors' write-ups of observational comments; authors' analytical memos on quotes and observations; and authors' field notes of observations and document analysis.

(Appendix C cont.)

Task (cont.)	Description (cont.)
Coding scheme	Details of the coding process are included in this manuscript in the “Methods” section, Figures 2 and 3, and Appendices C and D. Figures 2 and 3 show our case study methodology and data analysis process respectively. Appendices C and D further describe the open and axial coding with examples of these codes. NVivo was used to create coding nodes to assign open codes, prepare analytical memos with reflective commentary, and assign axial codes that further distilled the open codes into the core emerging themes from the data. The coding process ended when theoretical saturation occurred.
Validity	Wohlin’s (2012) validity framework was used to evaluate the study’s validity. Triangulation of data collection was used to analyse and compare interview, observation, and document data for the team studied. Data was collected at different points over time, and none of the authors were members of the ASD team. Authors explained all terms to the ASD team members prior to interviews, 2 researchers interpreted the data, and findings were reviewed by a senior agile practitioner. We also created an audit trail to show the process we followed, examples of open and axial coding, and diagrammatic illustrations of our case study methodology and data analysis process.
Case study overview	We provide a Table 1 with a case study profile to outline the team composition, experience levels, type of project, and end user for the project. We include a Table 2 that provides a data collection profile of individual interview and focus group compositions, interviews conducted, and meetings observed. Figure 2 provides our case study methodology, and lists all the data sources used, e.g. interviews, focus groups, supplemental documentation, and project photographs that were analysed.
Research report	We developed a scientific report with the relevant literature studied and used (to put theory in scholarly context); a detailed description of the case study context, methodology, and phenomena; an overview of our methods describing the research design, sampling, data collection, and data analysis (including coding process); the results from the data to answer the original research questions; a description of the grounded theory approach with examples of direct quotes capturing people’s personal perspectives and experiences; graphic displays (e.g. tables and figures); limitations and contributions of the study; implications for research and practitioners; recommendations for future research; references; appendices (e.g. interview guide and examples of open and axial coding).

APPENDIX D

Open and Axial Coding Process Examples with Analytical Memo in Conversation Squares and Participant Quotes

Data Coded: Examples of Participants' Words and Author Observations

Open Codes

Axial Codes

- "The SM decides the issues to address ...stories to handle...and any bugs to fix".
- "People defer to me because I'd be the senior QA".
- The SM "suggests somebody take a task".
- "The SM decides the initial task and gains consent from the corresponding developer or QA".
- "When I do raise concerns, it often falls on deaf ears as senior members don't do much about it so I don't usually get involved in decisions".
- Observation of senior members deciding how to develop new functionality components during SEM without consultation of all members.
- "The BA knows what's important for the client and the SM knows how to do it technically so they tell us what to do first."
- "The BA will sometimes speak to the client for consent, but for me, he [the BA] is the business owner."
- Meeting observations showed no client interaction; the team only interacted with the BA for the business perspective.
- Observation of timing problems continuing when junior members give inaccurate estimates.
- "I give a big estimate because the senior team members revise it anyway so I never have a good estimate".
- "I just give 2 days [for my estimates] because you really don't understand what is involved in dealing with a task, and it gives you time to figure it out."

Not all developers are involved in decisions

BA prioritises with SM; customer never involved with team (Customer Collaboration decision characteristic with decision process challenges).

There is little customer interaction in decisions

Junior team members decide estimates that are inaccurate

Senior team members override decisions or make decisions themselves

The axial code summarises the aspect that a senior team member overrides decisions across each of the open code categories of decisions.

These have issues across decision characteristics).

(Appendix D cont.)

Data Coded: Examples of Participants' Words and Author Observations

Open Codes

Axial Codes

- “We’ll work out whether a bug or an issue arose in the previous sprint needs to be fixed as a higher priority than doing any forward development”.
- “Then we would decide the resource thing”.
- “They would schedule how many days they think it would take considering their knowledge of the area”.
- Observations of decisions during the IPM.

Answers what decisions do you make during specific meetings? (Interactions over Process decision characteristic with decision process & intelligence challenges).

Similar task decisions for planning an iteration’s work each iteration

Decisions are repeated as we see IPMs giving same tasks to same people (Interactions over Process decision characteristic with decision process, intelligence & quality challenges).

- “[For tasks and estimation, the team member with] good experience in the particular area...normally, the first person picked is based on experience”.
- Someone decides to take the task if: “I did something like this before, I know how to do it...I can do this quick[ly]”.
- Observations of IPM showed tasks reallocated to the person who worked on a related task previously.

Experience is involved in repeated decisions and for deciding tasks.

Same people take similar tasks without considering other options, e.g. other people for the task or other tasks for that person

Non-compensatory decision process is used

Axial code summarises the aspect across open codes that decisions are not examining all relevant information.

- “If we delivered good, working functionality on time, then we made good decisions”.
- “Decision quality is high if decisions were made quickly”.
- “We know decisions are good if they save us time”.
- “This is one of the most difficult things is measuring the quality of your decisions because often it’s intangible and we have no standard way of knowing if we made good decisions”.

These examples show what participants said were good decisions.

Team focused on whether developed functionality worked and time to make decisions for determining if a decision was good.

Characteristics of good decisions made by ASD team

- “[We made] bad decisions when things get developed incorrectly”.
- “If a decision takes up more time, if there’s a lot of time wasted on it, then it was a bad decision...you’re wasting time or else you’re saving time”.
- “The quality of our decisions is based on an unclear mix of customer and team satisfaction”.

These examples show what participants said were bad decisions.

Team focused on mistakes in development and low satisfaction for determining if a decision was bad.

Characteristics of bad decisions made by ASD team

No standard measures of decision quality are used by the ASD team

Axial code summarises the aspect across open codes that decision quality has multiple anecdotal measures.